12TH CHEMISTRY · CHAPTER 17 · CHAPTER TEST

Group 2 Elements

Chapter-wise test: 6 MCQs, 5 of 6 short questions, 1 long question

Class: 12th (HSSC Part II)	**Subject:** Chemistry	**Total Marks:** 26
Chapter: 17	**Time Allowed:** 45 minutes	**Name:** _____

OBJECTIVE PART (6 × 1 = 6)

Q.1 Choose the correct answer.

1. Which Group 2 element does not react with water or steam?

(a) Barium (b) Magnesium
(c) Calcium (d) Beryllium

2. Which Group 2 carbonate is the most thermally stable?

(a) $BeCO_3$ (b) $BaCO_3$
(c) $CaCO_3$ (d) $MgCO_3$

3. Which Group 2 sulphate is the least soluble in water?

(a) $BaSO_4$ (b) $MgSO_4$
(c) $CaSO_4$ (d) $BeSO_4$

4. The coordination number of Be in its hydrated ion is:

(a) 2 (b) 6
(c) 4 (d) 8

5. Magnesium is extracted from seawater by the:

(a) Haber process (b) Dow process
(c) Contact process (d) Solvay process

6. "Milk of Magnesia" is:

(a) $MgSO_4$ (b) MgO
(c) $Mg(OH)_2$ (d) $MgCO_3$

SUBJECTIVE PART — SHORT QUESTIONS (5 × 2 = 10)

Q.2 Write short answers to any FIVE (5) questions.

i. Why does the reactivity of Group 2 elements increase down the group?
ii. Write equations for the reaction of magnesium with cold water and with steam.
iii. State two differences between Group 1 and Group 2 metals.
iv. Why is $MgCO_3$ thermally less stable than $CaCO_3$?
v. What is a complex ion? Give one example from Group 2.
vi. Why is $BaSO_4$ safe to use in "barium meals"?

LONG QUESTION (2 × 5 = 10)

Q.3 Answer both parts.

(a) Explain the trends in solubility of Group 2 hydroxides and sulphates.
(b) Describe the extraction of magnesium from seawater by the Dow process.

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